

DATA ANALYTICS PORTFOLIO

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# HR ANALYTICS ATTRITION ANALYSIS

*SQL Query Library + Power BI Dashboard*

PREPARED BY

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## 1. Project Overview

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This project analyzes employee attrition across 1,480 employee records, using a 30-query SQL library to compute headcount, compensation, tenure, and satisfaction metrics, and a two-page Power BI dashboard (“Attrition Overview” and “Attrition Drivers and Insights”) to present them to an HR stakeholder with Department and Job Role slicers.

As with the other query-to-dashboard projects in this portfolio, the SQL layer and the dashboard are meant to be auditable against each other. This one turned up more than the usual number of mismatches — most notably, several chart titles that were reused across the two dashboard pages but wired to different underlying fields. Section 10 documents each one with the evidence used to identify it, rather than silently relabeling the charts.

## 2. Objectives

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- Establish headcount and attrition fundamentals: total employees, attrition count, and the overall attrition rate.
- Break attrition down by department, job role, gender, age group, marital status, and education field to find where it concentrates.
- Examine compensation and tenure patterns — income by department and role, average age, average tenure — as context for retention.
- Test known attrition risk factors: overtime work, distance from home, number of prior companies, and years since last promotion.
- Package the results into a two-page, slicer-driven Power BI dashboard split into an executive overview and a drivers/insights deep-dive.

## 3. Dataset & Schema

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The data lives in a single table, HR\_Analytics, with one row per employee and 37 columns spanning demographics, compensation, role history, and satisfaction survey scores. The column set (AgeGroup, SalarySlab, EmployeeCount, Over18, StandardHours alongside the standard IBM-style attrition fields) matches the well-known public IBM HR Analytics Employee Attrition dataset this project is modeled on.

```
CREATE TABLE HR_Analytics (
  EmpID          VARCHAR(20) PRIMARY KEY,
  Age            INT,   AgeGroup      VARCHAR(20),
  Attrition      VARCHAR(10),
  Department     VARCHAR(50),
  JobRole        VARCHAR(100), JobLevel INT,
  Gender         VARCHAR(10), MaritalStatus VARCHAR(20),
  MonthlyIncome  INT,   SalarySlab    VARCHAR(20),
  OverTime       VARCHAR(10),
  YearsAtCompany INT,   YearsSinceLastPromotion INT,
  JobSatisfaction INT,   EnvironmentSatisfaction INT,
  WorkLifeBalance INT,   PerformanceRating      INT,
  ... -- 37 columns total; see HR_Analytics.sql for the full DDL
);
```

### Data cleaning step

Before the analytical queries, the script drops the EmpID primary key and de-duplicates rows using `ROW_NUMBER() OVER (PARTITION BY EmpID ORDER BY ctid)`, keeping only the first physical row per employee — a solid, reusable Postgres dedup pattern (ctid confirms the target dialect is PostgreSQL). One gap: the primary key constraint is dropped but never re-added after cleanup, which leaves EmpID unenforced as a unique key going forward. Worth adding `ALTER TABLE hr_analytics ADD PRIMARY KEY (EmpID);` back in once de-duplication is confirmed clean.

**Note on redundant columns:** EmployeeCount, Over18, and StandardHours are constant across every row in the source dataset this schema mirrors (always 1, 'Y', and 80 respectively) and carry no analytical signal. Likewise, MonthlyRate, DailyRate, and HourlyRate are known in the source dataset to be independently randomized and uncorrelated with MonthlyIncome — worth confirming before using them in any compensation analysis, rather than assuming they're consistent pay-rate conversions of each other.

## 4. Methodology

The 30-query library is organized into four tiers, moving from headcount fundamentals to increasingly specific risk-factor analysis:

- **Headcount & Attrition Fundamentals (Q1–Q9)** — totals, overall attrition rate, and attrition broken down by department, job role, gender, and age group.
- **Compensation & Demographics (Q10–Q13)** — average monthly income by department and role, the top 10 highest earners, and average age by department.
- **Work Conditions & Life Factors (Q14–Q18)** — overtime prevalence and its attrition rate, attrition by marital status, average commute distance for leavers, and job-hopping history.

- **Tenure, Development & Satisfaction (Q19–Q30)** — working years by role, attrition by education field / job level / salary slab, satisfaction and work-life-balance scores, promotion gaps, training frequency, and the top 5 highest-attrition job roles.

#### Design choice

Every rate calculation in the library uses the same pattern — `ROUND(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2)` — rather than a separate `COUNT(*)` FILTER or two-pass query. Consistent, portable, and easy to eyeball-audit across all 10+ rate queries in the script.

## 5. Key Performance Indicators

Headline metrics from Q1–Q3 and the dashboard's top-row summary cards:

|                                   |                                  |                                     |
|-----------------------------------|----------------------------------|-------------------------------------|
| TOTAL EMPLOYEES<br><b>1,480</b>   | ACTIVE<br><b>1,242</b>           | INACTIVE (ATTRITION)<br><b>238</b>  |
| ATTRITION RATE<br><b>16.08%</b>   | AVG. TENURE (YRS)<br><b>7.01</b> | AVG. MONTHLY SALARY<br><b>6.50K</b> |
| AVG. SALARY HIKE<br><b>15.21%</b> | MAX. SALARY HIKE<br><b>25%</b>   | "HIGH RISK" COUNT<br><b>31</b>      |

1,242 active + 238 inactive = 1,480 total, and  $238 \div 1,480 = 16.08\%$  — both check out exactly against Q1–Q3. The **"High Risk Count" of 31**, however, has no corresponding query anywhere in the 30-query library — its risk-scoring logic (whatever combination of overtime, satisfaction, and tenure it uses) exists only as an undocumented Power BI measure. See Section 10.

## 6. Dashboard Overview

The report is split across two tabs. "Attrition Overview" leads with headcount and salary KPIs and a "Who is leaving?" / "Where are they leaving from?" narrative structure.



Figure 1 — Attrition Overview tab, Power BI Desktop

“Attrition Drivers and Insights” repeats the same KPI row, then re-uses the same three chart titles (Age Group, Marital Status, By Overtime, Tenure vs Attrition, Promotion v/s Attrition) for a second, differently-scoped set of visuals, closing with two narrative insight panels.



Figure 2 — Attrition Drivers and Insights tab, Power BI Desktop

That title reuse is exactly what surfaces the labeling issues in Section 10 — comparing the two tabs side by side is what makes them visible.

## 7. Who Is Leaving — Attrition Demographics

Attrition rate by age group (Q9) shows a clear, monotonic pattern: attrition falls steadily as employees get older.

| Age Group | Attrition Rate |
|-----------|----------------|
| 18–25     | 35.77%         |
| 26–35     | 18.99%         |
| 55+       | 17.02%         |
| 46–55     | 11.84%         |
| 36–45     | 9.13%          |

Attrition rate by marital status (Q16) follows the same shape as most datasets of this kind — single employees leave at more than double the rate of married or divorced employees:

| Marital Status | Attrition Rate | Share of All Leavers |
|----------------|----------------|----------------------|
| Single         | 25.37%         | 52.74%               |
| Married        | 12.37%         | 25.72%               |
| Divorced       | 10.37%         | 21.55%               |

Attrition rate by gender — shown on the dashboard under the misleading title “By Overtime” (see Section 10) — confirms the insight text on the dashboard's second tab: male employees leave at a meaningfully higher rate than female employees.

| Gender | Attrition Rate | Share of Workforce |
|--------|----------------|--------------------|
| Male   | 16.99%         | 53.57%             |
| Female | 14.72%         | 46.43%             |

## 8. Compensation, Tenure & Department Benchmarks

Department-level attrition rate — shown on the dashboard under the title “Promotion v/s Attrition” on the Overview tab, but matching Q5's grouping and ordering exactly:

| Department             | Attrition Rate |
|------------------------|----------------|
| Sales                  | 20.67%         |
| Human Resources        | 19.05%         |
| Research & Development | 13.75%         |

Education-field attrition rate — shown on the Overview tab under the title “Tenure vs Attrition” (again, see Section 10) — is led by Human Resources and Technical Degree backgrounds:

| Education Field  | Attrition Rate |
|------------------|----------------|
| Human Resources  | 25.93%         |
| Technical Degree | 24.24%         |
| Marketing        | 22.36%         |
| Life Sciences    | 14.66%         |
| Medical          | 13.40%         |
| Other            | 13.25%         |

Q10–Q13 provide the compensation side of the picture: average monthly income by department and by job role, the top 10 highest earners, and average age by department. None of these are visualized on either dashboard tab — they're available in the query library as deeper-dive material for a compensation-equity review, but aren't part of the current dashboard narrative.

## 9. Tenure, Promotion & Career Growth

The Drivers and Insights tab's genuinely-tenure-based chart (also titled “Tenure vs Attrition”) shows attrition is heavily concentrated in the first two years — nearly 30% of employees with 0–2 years' tenure who are in this attrition-rate calculation have left, dropping off sharply after year 3:

| Years at Company | Attrition Rate |
|------------------|----------------|
| 0–2              | 29.82%         |
| 31–40            | 25.00% *       |
| 3–5              | 13.90%         |
| 6–10             | 12.17%         |
| 21–30            | 8.00%          |
| 11–20            | 6.63%          |

\* The 31–40-year bucket's 25.00% rate is almost certainly built on a very small employee count (few people at any company reach 31–40 years of tenure) — worth showing the underlying n for that bucket before quoting this figure, since a small denominator makes the rate unstable.

The corresponding promotion-gap chart (“Promotion v/s Attrition” on the Drivers tab) shows a similar early-tenure pattern: employees who haven't been promoted in 6–10 years show the highest attrition rate, though the low end (0–2 years since last promotion) is close behind — plausibly newer hires who haven't had time for a first promotion yet, rather than employees being overlooked.

| Years Since Last Promotion | Attrition Rate |
|----------------------------|----------------|
| 6–10                       | 18.00%         |
| 0–2                        | 16.91%         |
| 11–15                      | 12.12%         |
| 3–5                        | 10.13%         |

Q26 lists individual employees who haven't been promoted in over 5 years — a useful worklist for a manager follow-up, complementing the bucketed rate view above with names rather than aggregates.

## 10. Data Quality & Dashboard Labeling Notes

This dashboard's most significant issue isn't a data problem — it's a labeling problem. Both tabs reuse the same five chart titles (Age Group, Marital Status, By Overtime, Tenure vs Attrition, Promotion v/s Attrition), which is a reasonable design choice on its own, but several of those titles are attached to different underlying fields depending on which tab you're looking at. The table below documents each mismatch and the evidence used to identify it:

| Tab      | Displayed Title         | Actual Data Shown                            | Backing Query                               |
|----------|-------------------------|----------------------------------------------|---------------------------------------------|
| Overview | By Overtime             | Gender (Male / Female)                       | Q8 (counts only)                            |
| Overview | Tenure vs Attrition     | Attrition rate by Education Field            | Not in library (Q20 is counts only)         |
| Overview | Promotion v/s Attrition | Attrition rate by Department                 | Q5 — matches exactly                        |
| Drivers  | By Overtime             | Gender (identical to Overview tab)           | Q8 (counts only)                            |
| Drivers  | Marital Status          | Attrition rate by Job Satisfaction (1–5)     | Not in library                              |
| Drivers  | Tenure vs Attrition     | Attrition rate by Years-at-Company band      | Not in library                              |
| Drivers  | Promotion v/s Attrition | Attrition rate by Years-Since-Promotion band | Not in library (Q26 lists individuals only) |

### How this was verified

The “By Overtime” mismatch was confirmed by matching its 46.43%/53.57% split and 14.72%/16.99% rates exactly against the dashboard's own “male employees show a higher attrition rate” insight text — that statement only makes sense if the chart is Gender, not OverTime (Yes/No). The Tenure/Promotion swaps were confirmed by comparing the two tabs directly: the Overview tab's “Tenure vs Attrition” shows education-field category labels (Human Resources, Life Sciences, Medical, etc.), while the Drivers tab's chart of the same name shows genuine year-range labels (0–2, 3–5, 6–10...). The same pattern holds for “Promotion v/s Attrition.” This reads as charts being duplicated from one tab to build the other, with the underlying field swapped but the title left unchanged.

Beyond the labeling, three smaller items are worth a look before this goes in front of a stakeholder:

- **Salary card mislabel** — the Avg. Monthly Salary KPI card pairs “Max. Salary: 19999” with “Min. Hike: 11,” mixing a salary figure with a hike figure. The separate Avg. Salary Hike (%) card already has its own correctly-paired “Max. Hike: 25 / Min. Hike: 11.” The salary card's second sub-stat most likely should read “Min. Salary,” not “Min. Hike.”
- **Undefined “High Risk Count”** — as noted in Section 5, no query in the library defines what makes an employee “high risk.” The 31-employee count is real but its rule (some combination of overtime, low satisfaction, tenure, etc., presumably) isn't documented anywhere reproducible.
- **Q15 has no dashboard visual** — Q15 (attrition rate among overtime workers) is the one query that would correctly populate a real “By Overtime” chart, comparing OverTime = Yes vs. No. It's computed in SQL but never surfaced on either dashboard tab — the chart titled “By Overtime” shows Gender instead, so this comparison is currently invisible to a dashboard viewer.

## 11. Tools & Technical Skills Demonstrated

- SQL: window-function de-duplication (ROW\_NUMBER partitioned by EmpID), a consistent CASE-based rate-calculation pattern reused across 10+ queries, multi-column GROUP BY, and ranked filtering with ORDER BY / LIMIT.
- Power BI: multi-tab report design, bookmark-style tab navigation, cross-filtering slicers (Department, Job Role), donut and bar visuals with dual metrics (rate + share) in a single label.
- Analytical thinking: auditing a dashboard against its own query library rather than taking chart titles at face value, cross-referencing two dashboard tabs to catch a field/title mismatch that neither tab alone would reveal, and recognizing dataset-level quirks (constant columns, randomized rate fields) before they contaminate an analysis.

## 12. Conclusion & Recommendations

Attrition is concentrated in a clear, coherent profile: young (18–25), single, early-tenure (0–2 years) employees in Sales or HR-education backgrounds, with male employees leaving at a somewhat higher rate than female employees. That's a specific, actionable retention target rather than a diffuse company-wide problem.

- Fix the chart-title/field mismatches in Section 10 before this dashboard is shared externally — at minimum, retitle the four affected charts and add the missing “By Overtime (Yes/No)” visual that Q15 already supports.
- Document the “High Risk Count” scoring rule (or rebuild it as a transparent SQL query) so the 31-person figure is reproducible and auditable, not just a black-box Power BI measure.
- Prioritize a first-two-years retention program — onboarding, early mentorship, or a first-year check-in — given that both the tenure-band and promotion-gap charts point to the same early-career window as the highest-risk period.